Through Zero and Beyond – One Step at a Time

Decimal Dina is on a rescue mission in the Ice Crystal Caverns, where temperatures drop below zero, and mysterious number patterns are frozen in blocks of ice. Help her navigate by solving these number puzzles!

Questions

1. Dina enters the first cavern where the temperature is **–7°C.** Outside, it was **5°C.** What is the difference in temperature between the two places?

a) 12°C

b) –12°C

c) –2°C

d) None of the above

2. On Day 1, the temperature was **–2°C.** On Day 2, it dropped **5** more degrees. What was the temperature on Day 2?

a) –7°C

b) 3°C

c) –3°C

d) None of the above

3. A frozen map shows: –8, –3, 0, 4, –6

Match the numbers to their positions from coldest to warmest:

a) –8	**Warmest**
b) –6	**Middle value**
c) –3	**Second coldest**
d) 0	**Coldest**
e) 4	**Second warmest**



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4. Dina finds a scale with marks: –10, –5, 0, 5, 10.

Match the situation to the correct number:

a) 10 floors above	**-10**
b) 5 floors above ground	**-5**
c) 5 meters below sea	**0**
d) Midpoint between –10 and 10	**5**
e) Very cold freezer temperature	**10**

5. A crystal clue reads:

–5 is greater than –2.

True or False?

6. Dina checks a mountain cave where the outside temperature is **–3°C.** The temperature inside is **4°C warmer.** What is the temperature inside the cave?

7. Dina finds a thermometer reading.

Fill in the blank:

The temperature dropped from 3°C to –4°C. The total drop is _____ degrees.

8. A frost dragon guards a puzzle:

Fill in the blanks:

–3 is _____ than –7

–1 is _____ than 0

(Options: greater, less)



9. Inside the Frosty Library, Dina finds these facts:

a) $-1 < 0$

b) $-9 > -3$

c) $-6 = -6$

Which statements are correct?

a) a and b only

b) b and c only

c) a and c only

d) All of the above

10. Dina's sled starts at **–4 meters** below sea level and climbs to **2 meters** above.

Fill in the blank:

The sled climbed _____ meters in total.

Thanks for helping Dina thaw out the mysteries of negative numbers!

She's now one step closer to the Frozen Formula of Fractions!

